

CHAPTER 2

Resource Mobilization in Indian Economy: Sources of financial resources for public and private sectors - budgetary resources - tax revenue and non-tax revenue - public debt : market borrowings, loans and grants etc., external debt from multilateral agencies - foreign institutional investment and foreign direct investment - desirability and consequences of utilizing different sources - Monetary and fiscal policies - financial markets and institutions of developmental finance - investment in industries and infrastructure projects - Physical resources - Energy resources

INTRODUCTION:

Resource mobilization is a fundamental aspect of the Indian economy, encompassing the acquisition and allocation of financial, human, and physical resources to fuel economic growth, development, and sustainability. In this lesson, we will focus on the sources of financial resources for both the public and private sectors within the Indian economy, examining various aspects such as budgetary resources, tax and non-tax revenue, public debt, external debt, foreign investment, and their implications. Additionally, we will touch upon the role of monetary and fiscal policies, financial markets, developmental finance institutions, and the significance of physical resources, particularly energy resources, in the context of resource mobilization.

Budgetary resources play a pivotal role in the Indian economy, serving as the foundation for the government's fiscal planning and expenditure management.

Budgetary Resources in India:

Union Budget:

The Union Budget is the annual financial statement of the Indian government. It outlines the government's anticipated revenues and planned expenditures for the fiscal year. It consists of two key components:

- a. Revenue Budget:** This part of the budget deals with the government's revenue receipts and expenditures that do not create physical or financial assets. It includes tax and non-tax revenue, grants, and subsidies.
- b. Capital Budget:** The capital budget deals with capital receipts and capital expenditures. Capital receipts include borrowings and disinvestment proceeds, while capital expenditures are used to create physical or financial assets, such as infrastructure projects and investments.

Significance of Budgetary Resources:

- **Economic Planning:** The budgetary resources are at the heart of India's economic planning. They provide a framework for allocating funds to various sectors and programs, supporting the government's development goals.
- **Resource Allocation:** The allocation of resources through the budgetary process influences the direction of economic growth. Prioritizing sectors like healthcare, education, infrastructure, and agriculture can foster inclusive development.
- **Fiscal Discipline:** Budgetary resources help in maintaining fiscal discipline. A balanced budget ensures that government expenditure does not exceed its revenue, preventing excessive borrowing and fiscal deficits.
- **Revenue Generation:** Tax revenue, a crucial component of budgetary resources, funds various government initiatives. Increasing tax compliance and expanding the tax base are essential to boost revenue generation.

Implications for the Indian Economy:

- **Economic Growth:** Budgetary resources are essential for financing development projects, infrastructure, and social programs. Proper allocation and utilization can stimulate economic growth and job creation.
- **Fiscal Deficit:** A high fiscal deficit, caused by excessive spending or inadequate revenue, can strain the economy. Managing budgetary resources effectively is crucial to keep the deficit in check.
- **Inclusivity:** Allocating budgetary resources to sectors like healthcare and education can improve access to basic services and promote social inclusivity, reducing disparities in society.
- **Infrastructure Development:** Capital expenditures from budgetary resources are instrumental in building infrastructure, which is critical for economic development and competitiveness.
- **Investor Confidence:** A well-balanced budget with prudent fiscal management can boost investor confidence, both domestic and foreign, fostering economic stability.
- **Monetary Policy Interaction:** The government's fiscal policies, driven by budgetary resources, interact with the monetary policies of the central bank to shape overall economic conditions.

In conclusion, budgetary resources in the Indian economy are the cornerstone of fiscal planning and governance. They determine the government's ability to fund critical programs, infrastructure development, and social services. Effective management of these resources is vital for achieving economic growth, fiscal discipline, and inclusivity.

Tax and non-tax revenues

These are essential components of government income in any country. In the Indian context, these revenues play a crucial role in financing public expenditure, driving economic growth, and achieving developmental goals. This essay explores the concepts, significance, and challenges associated with tax and non-tax revenue in the Indian economy.

Tax Revenue

Tax revenue refers to the funds collected by the government from various taxes imposed on individuals, businesses, and economic activities. In India, tax revenue comprises two main categories: direct taxes and indirect taxes.

Direct Taxes:

- **Income Tax:** Income tax is a progressive tax levied on the income earned by individuals and businesses. It is a significant source of revenue for the Indian government. The tax rates vary depending on income levels.
- **Corporate Tax:** Corporate tax is imposed on the profits of companies and businesses. It plays a vital role in generating revenue and encouraging economic activities.
- **Wealth Tax (abolished in 2015):** It was levied on the net wealth of individuals. Its abolition was aimed at simplifying the tax structure.

Indirect Taxes:

- **Goods and Services Tax (GST):** GST is a comprehensive indirect tax levied on the supply of goods and services. It replaced a complex system of multiple indirect taxes, streamlining tax collection and improving compliance.
- **Customs Duty:** It is a tax imposed on the import and export of goods. It serves not only as a revenue source but also as a tool for regulating international trade.
- **Excise Duty:** Excise duty is levied on the manufacture of goods, such as alcohol, tobacco, and petroleum products. It contributes significantly to government revenue.

Non-Tax Revenue

Non-tax revenue refers to the income generated by the government through sources other than taxes. It encompasses a diverse range of income streams:

- **Dividends and Profits:** The government earns dividends and profits from public sector enterprises in which it holds a stake. These enterprises include companies in sectors like oil, telecommunications, and banking.
- **Interest and Fees:** The government earns interest on loans and fees from various services, such as licenses, permits, and fines.
- **Grants-in-Aid:** The central government receives grants from international organizations, foreign governments, and other domestic entities to finance specific projects and programs.
- **Monetization of Assets:** The government can generate revenue by selling or leasing public assets, such as land, buildings, or spectrum.

Significance of Tax and Non-Tax Revenue

- **Fiscal Stability:** Tax revenue provides a stable and consistent source of income for the government, enabling it to meet its fiscal obligations and maintain economic stability.
- **Development Finance:** Tax and non-tax revenues are vital for financing development projects, social welfare programs, and infrastructure development, contributing to economic growth and poverty reduction.
- **Economic Regulation:** Indirect taxes like GST and customs duties serve not only as revenue sources but also as tools for regulating economic activities and international trade.
- **Resource Allocation:** Tax policy can be used to influence resource allocation, incentivize certain behaviours, and redistribute wealth.
- **Reducing Income Inequality:** Progressive tax policies can help reduce income inequality by taxing higher-income individuals and corporations at higher rates.

Challenges and Concerns

- **Tax Evasion and Avoidance:** Tax evasion and avoidance remain significant challenges, leading to revenue leakage and inequity in the tax system.
- **Complexity:** India's tax system can be complex and cumbersome, posing challenges for both taxpayers and tax administrators.
- **Informal Economy:** A substantial part of the Indian economy operates in the informal sector, often escaping taxation.
- **Tax Burden:** Imposing a high tax burden can discourage economic activities and hinder economic growth.

Conclusion

Tax and non-tax revenues are the lifeblood of government finances in India. They enable the government to fulfil its responsibilities, promote economic growth, and address social and developmental needs. Effective tax administration, simplified tax systems, and efforts to reduce tax evasion are essential to enhance revenue collection and achieve fiscal sustainability. Balancing the tax burden while maximizing non-tax revenue sources is critical for fostering economic growth and equitable development in India.

Public debt

It is a significant component of a nation's financial framework, representing the accumulated borrowing by the government to finance its various expenditures, ranging from infrastructure development to social welfare programs. In India, as in many other countries, public debt plays a vital role in economic management and policy formulation. This essay explores the concept, types, significance, challenges, and implications of public debt in the Indian context.

Concept of Public Debt

Public debt, often referred to as government debt or sovereign debt, is the total amount of money borrowed by the government from domestic and international sources. It is used to cover budget deficits, finance

development projects, and meet other financial obligations. Public debt consists of both internal debt (borrowings within the country) and external debt (borrowings from foreign sources).

Types of Public Debt

- **Internal Debt:** This includes borrowings made by the government within the country. Internal debt is further classified into marketable and non-marketable debt.
- **Marketable Debt:** These are government securities that can be traded in the financial markets. They include Treasury bills, government bonds, and other marketable instruments.
- **Non-Marketable Debt:** This debt comprises loans and deposits that are not tradable in the financial markets.
- **External Debt:** External debt refers to loans taken by the government from foreign sources. It includes bilateral and multilateral loans, commercial borrowings, and loans from international financial institutions.

Significance of Public Debt

- **Financing Development:** Public debt is often used to finance critical infrastructure projects, social welfare programs, and economic development initiatives, which may not be feasible through tax revenues alone.
- **Economic Stability:** It helps in stabilizing the economy during economic downturns by enabling the government to continue spending and supporting economic growth.
- **Investor Confidence:** Well-managed public debt can enhance investor confidence by demonstrating fiscal responsibility and commitment to debt servicing.
- **Monetary Policy Tool:** Public debt issuance can be used to implement monetary policy objectives, including managing liquidity in the financial markets.
- **Capital Formation:** It promotes capital formation by channelling savings into productive investments.

Challenges and Concerns

- **Debt Sustainability:** A key challenge is ensuring that the level of public debt remains sustainable over the long term. Excessive borrowing can lead to debt servicing difficulties and fiscal crises.
- **Interest Payments:** A significant portion of government expenditure is allocated to servicing the interest on public debt, which can limit resources available for other critical areas.
- **Crowding Out:** Excessive public debt can crowd out private investment by raising interest rates and reducing the availability of credit to the private sector.
- **External Debt Management:** Managing external debt requires careful consideration of foreign exchange risks, as fluctuations in exchange rates can impact debt servicing obligations.

Implications of Public Debt

- **Fiscal Policy:** Public debt affects fiscal policy decisions, including taxation, government spending, and deficit management.
- **Monetary Policy:** It influences the implementation of monetary policy by impacting interest rates and money supply.
- **Economic Growth:** Prudent public debt management can contribute to economic growth by financing essential projects and programs.

Conclusion

Public debt is a critical financial instrument that allows the government to fulfil its fiscal obligations and promote economic development. However, it must be managed carefully to avoid the risks associated with excessive debt levels. Effective debt management, transparency, and accountability are essential for ensuring that public debt remains sustainable and contributes positively to the overall well-being of the economy. In

the Indian context, public debt management plays a crucial role in supporting the country's economic growth and development aspirations.

Foreign Institutional Investment (FII) and Foreign Direct Investment (FDI)

These are critical components of India's economic landscape, attracting foreign capital to stimulate growth and development.

Distinctions between FII and FDI:

Foreign Institutional Investment (FII):

- **Portfolio Investment:** FIIs involve the purchase of financial assets such as stocks, bonds, and other securities in the Indian financial markets.
- **Short-Term:** FIIs are typically short-term investments, with investors seeking capital gains or dividends.
- **Low Control:** FIIs have no significant control over the companies in which they invest.

Foreign Direct Investment (FDI):

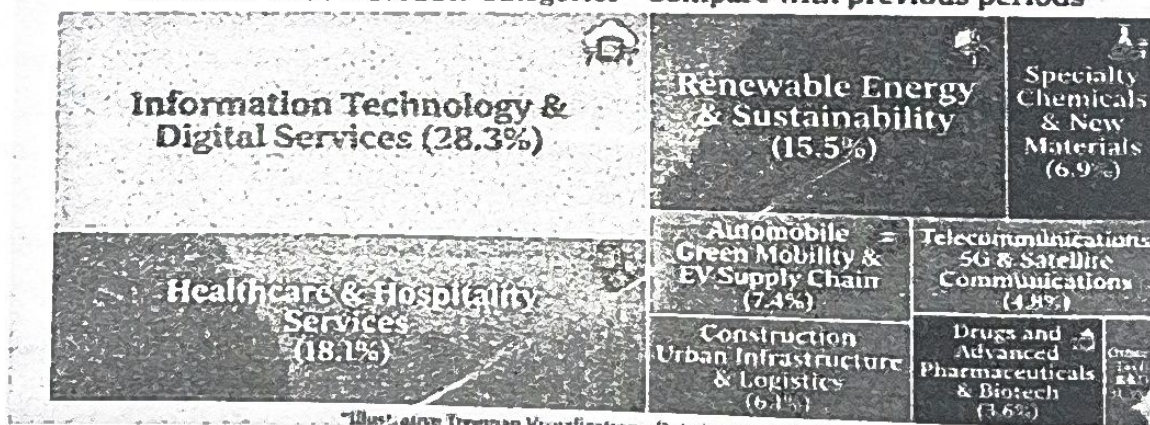
- **Long-Term Investment:** FDIs involve foreign entities establishing a physical presence in India, such as subsidiaries, joint ventures, or wholly-owned companies.
- **Strategic Control:** FDIs allow foreign investors to exert significant control and influence over the invested entities.
- **Long-Term Commitment:** FDIs represent a long-term commitment to the Indian economy.

Impact on the Indian Economy:

- **Economic Growth:** FDI contributes to sustained economic growth by creating jobs, enhancing productivity, and bringing new technologies and practices.
- **Portfolio Diversification:** FII provides an opportunity for portfolio diversification, which can contribute to financial market stability.
- **Foreign Exchange Reserves:** FII inflows increase foreign exchange reserves, bolstering India's economic stability.
- **Capital Inflows:** Both FII and FDI contribute to capital inflows, which can finance development projects, infrastructure, and reduce current account deficits.
- **Exchange Rate Volatility:** Fluctuations in FII can lead to exchange rate volatility, impacting import/export competitiveness.
- **Trend of FDI Inflows:** A line graph showcasing the trend of FDI inflows over the last decade, illustrating growth or fluctuations.

Sector-wise FDI Equity Inflows during April-September 2024 (or a subsequent representative period, e.g., Oct '24-Mar '25)

Indicative Data & Broader Categories - Compare with previous periods



*Illustrative Treemap Visualization - Data is representative

Current data: India's gross Foreign Direct Investment (FDI) inflows rose to a record high of \$94.53 billion in 2025-26 — up 17% from the previous year.

Impact on Economic Growth: A line graph demonstrating the correlation between FDI inflows and India's GDP growth rate over time.

Conclusion:

FII and FDI are essential drivers of India's economic growth and development. FDI fosters long-term commitment and strategic control, while FII provides liquidity and portfolio diversification. Both forms of foreign investment contribute to capital inflows, which can finance critical projects and stabilize the economy. However, prudent management is essential to mitigate risks associated with fluctuations in FII and to ensure that FDI contributes to sustainable growth and development.

Financial markets and institutions of developmental finance

There are crucial components of a nation's financial infrastructure. They play a pivotal role in the allocation of capital, mobilization of savings, and financing of economic development. In India, these institutions and markets have undergone significant transformation and expansion over the years.

Financial Markets in India

Financial markets in India encompass various segments, including capital markets, money markets, and derivatives markets. They serve as platforms for the trading of various financial instruments like stocks, bonds, currencies, and commodities.

Capital Markets: These markets facilitate the buying and selling of long-term financial instruments, primarily equities and bonds. The National Stock Exchange (NSE) and the Bombay Stock Exchange (BSE) are two prominent stock exchanges in India.

Money Markets: Money markets deal with short-term financial instruments. These include Treasury bills, commercial papers, and certificates of deposit. Money market instruments are essential for liquidity management by banks and financial institutions.

Derivatives Markets: Derivatives markets provide financial instruments like futures and options that derive their value from underlying assets. They offer risk management tools and opportunities for hedging and speculation.

Developmental Finance Institutions: Developmental finance institutions are specialized financial institutions created to promote and support specific sectors or activities critical for economic development. In India, these institutions have played a significant role in funding infrastructure, agriculture, and small and medium enterprises (SMEs).

Some notable developmental finance institutions include,

- **National Bank for Agriculture and Rural Development (NABARD):** NABARD focuses on rural and agricultural development by providing credit, refinancing, and development assistance to agriculture-related institutions and projects.
- **Small Industries Development Bank of India (SIDBI):** SIDBI supports the growth and development of SMEs by offering various financial and non-financial services, including credit guarantees and venture capital.
- **Infrastructure Development Finance Company (IDFC):** IDFC plays a pivotal role in financing infrastructure projects in sectors like transportation, energy, and urban development.

Importance and Functions

Financial markets and developmental finance institutions are essential for several reasons,

- **Capital Allocation:** They facilitate the efficient allocation of savings to productive investments, fostering economic growth.
- **Risk Management:** Derivatives markets allow businesses to manage risk by hedging against adverse price movements.

- **Promoting Inclusive Growth:** Developmental finance institutions focus on sectors often neglected by commercial banks, such as agriculture and SMEs, thus promoting inclusive economic development.
- **Infrastructure Financing:** Developmental finance institutions bridge the infrastructure funding gap by channelling long-term funds into crucial projects.
- **Liquidity and Stability:** Money markets provide liquidity to financial institutions and contribute to overall financial stability.

Challenges and Concerns

Despite their importance, financial markets and developmental finance institutions face challenges,

- **Regulatory Framework:** Maintaining an appropriate regulatory framework to ensure market integrity and investor protection is a constant challenge.
- **Access to Finance:** Some segments of society and businesses still face difficulties in accessing financial services and products.
- **Infrastructure Constraints:** Developing robust infrastructure for financial markets, especially in rural areas, remains a challenge.

Impact on the Indian Economy

Financial markets and developmental finance institutions have a profound impact on the Indian economy,

- **Economic Growth:** They channel funds into productive sectors, driving economic growth.
- **Inclusive Development:** Developmental finance institutions contribute to inclusive growth by supporting marginalized sectors and regions.
- **Investor Confidence:** Efficient financial markets and well-regulated institutions enhance investor confidence and attract foreign investment.
- **Infrastructure Development:** Developmental finance institutions are key players in India's infrastructure development.

Conclusion

In conclusion, financial markets and developmental finance institutions are integral to India's economic development. They facilitate capital allocation, risk management, and inclusive growth. While they have made significant strides, there is still room for improvement in terms of access, infrastructure, and regulatory framework. Ensuring their effective functioning is vital for India's sustained economic growth and development aspirations.

Investment in industries and infrastructure projects

It is a cornerstone of economic development in any nation. In the Indian context, these investments are essential drivers of growth, job creation, and enhanced competitiveness. This essay explores the significance of investment in industries and infrastructure, the challenges it faces, and its role in India's economic trajectory.

Examples of Impactful Investments

These flagship programmes represent major public investments aimed at reducing logistics costs, improving multimodal connectivity, boosting industrial competitiveness, and driving economic growth under the broader vision of *Atmanirbhar Bharat* and *Viksit Bharat@2047*.

1. Bharatmala Pariyojana

- **Launched:** October 2017 (approved by CCEA).
- **Nodal Ministry:** Ministry of Road Transport and Highways (MoRTH).
- **Objective:** Optimise freight and passenger movement by bridging critical gaps in the National Highway network through a corridor-based approach. Aims to reduce logistics costs (from ~13-14%

of GDP closer to global levels of 8–10%), improve inter-state connectivity (including border, coastal, and backward areas), and enhance economic corridors.

- **Phase-I Target:** ~34,800 km of highways (later rationalised; ~26,425 km awarded).
- **Key Components:**
 - Economic Corridors
 - Inter-corridors and Feeder Routes
 - National Corridor Efficiency Improvement (Including ring roads and decongestion)
 - Border and International Connectivity Roads
 - Coastal and Port Connectivity Roads
 - Expressways
- **Status (as of mid-2026):** Over 22,700 km constructed; projects for the rationalised length awarded. New sanctions under Phase-I discontinued since late 2023; focus on completion (revised target around 2027–28). Predominant use of Hybrid Annuity Model (HAM).
- **Impact:** Strengthens the Golden Quadrilateral and North-South/East-West corridors; supports multimodal logistics parks and industrial growth.

2. Sagarmala Programme

- **Launched:** 2015 (Cabinet approval March 2015; National Perspective Plan released 2016).
- **Nodal Ministry:** Ministry of Ports, Shipping and Waterways.
- **Vision:** Port-led development to reduce logistics costs for EXIM and domestic trade by leveraging India's ~7,500 km coastline and inland waterways. Promote coastal shipping and inland waterways over road/rail where feasible.
- **Five Pillars / Components:**
 1. Port Modernisation & New Port Development
 2. Port Connectivity Enhancement (road, rail, inland waterways)
 3. Port-led Industrialisation (Coastal Economic Zones, industrial clusters)
 4. Coastal Community Development (skills, livelihoods, fisheries, tourism)
 5. Coastal Shipping and Inland Waterways Transport
- **Status (as of 2026):** ~845 projects identified (estimated cost ~₹6.06 lakh crore); ~315 projects completed (investment ~₹1.56–1.57 lakh crore). Significant growth in coastal shipping traffic and cargo handling at major ports.
- **Impact:** Lowers transportation costs for bulk commodities, boosts export competitiveness through port-proximate manufacturing, generates employment in coastal areas, and supports multimodal integration. Sagarmala 2.0 extends the vision toward a larger maritime economy.

3. Dedicated Freight Corridors (DFC)

- **Implementing Agency:** Dedicated Freight Corridor Corporation of India Ltd. (DFCCIL).
- **Objective:** Create high-capacity, high-speed dedicated tracks exclusively for freight to decongest existing mixed-traffic railway lines, increase average freight speeds (from historical ~25 km/h toward 50–100 km/h), raise axle loads, enable double-stack container trains, and cut logistics costs and transit times.
- **Key Corridors:**
 - **Eastern DFC:** ~1,337 km (Ludhiana to Sonnagar/Dankuni area) — fully operational since ~2023–24.
 - **Western DFC:** ~1,506 km (Dadri to JNPT, Navi Mumbai) — fully commissioned by March 2026.
 - Combined length of the two main corridors: ~2,843 km.

- **Status:** Both Eastern and Western DFCs fully operational as of 2026. Additional corridors (East-West, North-South, East Coast, etc.) are under planning or early stages; some proposed under Hybrid Annuity Model.
- **Impact:** Significant reduction in transit times (especially coal and containers), higher rail share in freight, lower fuel consumption and emissions per unit of cargo, improved competitiveness of Indian industry, and better port-hinterland connectivity. Estimated substantial GDP contribution through logistics efficiency gains.

4. Industrial Corridors (with focus on Visakhapatnam-Chennai Industrial Corridor - VCIC)

- **Framework:** Developed under the National Industrial Corridor Development and Implementation Trust (NICDIT) / National Industrial Corridor Development Corporation (NICDC). Aim to create world-class infrastructure for manufacturing hubs along multimodal transport spines.
- **Visakhapatnam-Chennai Industrial Corridor (VCIC):**
 - Phase-I of the East Coast Economic Corridor (ECIC).
 - Length: Approximately 800 km along the eastern coast (primarily along NH-16 and the Kolkata-Chennai trunk railway).
 - States: Primarily Andhra Pradesh (extending influence toward Tamil Nadu/Chennai).
 - **Key Nodes:** Visakhapatnam, Chittoor (Srikalahasti-Yerpedu), Koppa (Kadapa), with others such as Machilipatnam and Donakonda identified.
 - **Support:** Multitranchise financing from Asian Development Bank (ADB) for infrastructure (roads, water, power, effluent treatment, etc.) and policy reforms. State-level Andhra Pradesh Industrial Corridor Development Authority involved.
 - **Status (2026):** Active implementation. Master plans completed/notified for priority nodes (e.g., Koppa); trunk infrastructure works progressing in several clusters (Rambilli, Nakkapalli, Naidupeta, etc.). Land and environmental clearances advanced for key parcels.
- **Broader Impact of Industrial Corridors:** Attract domestic and foreign investment in manufacturing, create high-quality jobs, promote sustainable and green industrial development, integrate with ports/highways/rail (including DFCs and Sagarmala), and reduce regional disparities along the east coast.

Significance of Investment in Industries and Infrastructure Projects

- **Economic Growth:** Investment in industries and infrastructure projects fuels economic growth by increasing productive capacity, facilitating trade, and creating jobs.
- **Job Creation:** These investments generate employment opportunities, particularly in labour-intensive industries and construction sectors, reducing unemployment and poverty.
- **Productivity Enhancement:** Infrastructure investments, such as roads, ports, and digital connectivity, improve supply chain efficiency, reducing logistics costs and enhancing productivity.
- **Competitiveness:** A robust industrial and infrastructure base enhances a country's competitiveness on the global stage, attracting foreign investment and fostering domestic entrepreneurship.
- **Social Welfare:** Investment in sectors like healthcare and education leads to improved social outcomes, better health services, and a more educated workforce.

Challenges in Investment

- **Funding Gap:** Financing large-scale infrastructure projects often requires substantial capital, leading to funding gaps. Both public and private investments are essential to bridge these gaps.
- **Regulatory Bottlenecks:** Cumbersome regulatory procedures, land acquisition issues, and environmental clearances can delay projects, increasing costs and deterring investors.

- **Political Uncertainty:** Frequent policy changes and political instability can discourage long-term investments.
- **Infrastructure Quality:** Despite significant investments, India faces challenges in maintaining the quality and sustainability of infrastructure assets.

Role of Government Policies

The government plays a pivotal role in encouraging investment in industries and infrastructure projects through various policies and initiatives,

- **Fiscal Incentives:** Tax breaks and financial incentives encourage private sector investments in industries and infrastructure.
- **Public-Private Partnerships (PPPs):** PPP models promote collaboration between the government and private sector, sharing risks and responsibilities in infrastructure development.
- **Ease of Doing Business:** Streamlining bureaucratic processes and improving the ease of doing business attract investments.
- **Investor Protection:** Ensuring the protection of investors' rights and assets instils confidence.

Examples of Impactful Investments

- **Golden Quadrilateral:** The construction of a network of highways connecting major cities has significantly improved connectivity and economic activity.
- **Make in India:** The government's "Make in India" initiative encourages foreign and domestic investments in various industries, promoting manufacturing and job creation.
- **Smart Cities Mission:** Investments in smart cities enhance urban infrastructure and promote sustainable urbanization.

Conclusion:

Investment in industries and infrastructure projects is a cornerstone of India's economic development. It fosters growth, job creation, and competitiveness while improving the quality of life for citizens. Addressing funding gaps, streamlining regulations, and providing a conducive investment environment are key challenges that need to be addressed. With a supportive policy framework and prudent management, India can continue to attract investments that drive its economic trajectory forward, meeting the aspirations of its people and enhancing its global standing.

Energy resources

These are the lifeblood of modern economies, driving industrialization, urbanization, and overall socio-economic development. They play a pivotal role in powering industries, transportation, agriculture, and households. In India, the significance of energy resources cannot be overstated, given the country's rapidly growing population and economy.

Importance of Energy Resources

- **Economic Growth:** Energy resources are the engine of economic growth, powering industries, manufacturing, and services sectors.
- **Infrastructure Development:** Adequate energy supply is vital for building and maintaining infrastructure, including roads, railways, and communication networks.
- **Quality of Life:** Access to reliable and affordable energy resources enhances the quality of life by providing lighting, heating, cooling, and powering electronic devices in households.
- **Agriculture:** Energy resources are crucial for modern agriculture, powering irrigation, machinery, and transportation of agricultural products.
- **Environmental Concerns:** Transitioning to cleaner and sustainable energy sources can help mitigate environmental issues, including air pollution and greenhouse gas emissions.

Energy Transition in India (Coal → Gas → Renewables → Green Hydrogen)
 India's energy transition follows a pragmatic, phased pathway shaped by energy security, rising demand, climate commitments (Net Zero by 2070), and economic development needs. The sequence **Coal → Natural Gas → Renewables → Green Hydrogen** reflects a shift from carbon-intensive baseload to cleaner, flexible, and eventually zero-carbon energy systems.

1. Current Energy Landscape (as of mid-2026)

- **Total installed power capacity:** ~549 GW.
- **Coal/Lignite:** Still the dominant source for actual generation (~70% during peak summer months; ~40–46% of installed capacity). Provides reliable baseload.
- **Renewables (including large hydro):** ~288–290 GW installed (solar ~162 GW leading, followed by wind ~57 GW and hydro ~57 GW). Non-fossil capacity (RE + nuclear) ~297 GW (~50%+ of total capacity).
- **Natural Gas:** Low share in power generation (<2–5% of capacity/generation) due to high prices, limited domestic supply, and infrastructure constraints.
- **Nuclear:** ~8–9 GW (small but stable).

Despite rapid RE capacity addition, coal remains critical for meeting peak and evening demand because of intermittency of solar/wind.

2. Stage-wise Transition Pathway

Stage 1: Coal Dominance (Present & Near-term)

- Backbone of India's power system due to abundant domestic reserves, low cost, and ability to provide 24×7 baseload.
- Challenges: High CO₂ emissions, air pollution, and international climate pressure.
- Mitigation measures: Efficiency improvement (supercritical/ultra-supercritical plants), coal gasification (National Coal Gasification Mission), and future Carbon Capture, Utilisation & Storage (CCUS).
- Coal will continue as a bridge fuel at least till the mid-2030s while RE + storage scales up.

Stage 2: Natural Gas as Transition Fuel

- Cleaner than coal (lower CO₂ and almost no particulate emissions).
- Role: Flexible peaking power, industrial feedstock, city gas distribution, and transportation (CNG/LNG).
- Constraints: High import dependence (LNG), price volatility, limited pipeline infrastructure, and low domestic production.
- Policy push: Expansion of city gas networks, PNG/CNG coverage, and gas-based power plants for balancing RE.
- Gas acts as a **bridge** between coal and full renewable systems but has not scaled as expected due to economics.

Stage 3: Renewables (Solar + Wind + Hydro + Storage) – Accelerating Phase

- Core of India's clean energy push.
- **Key Targets (Updated NDC / Panchamrit):**
 - 500 GW non-fossil capacity by 2030.
 - 50% of installed capacity from non-fossil sources by 2030.
 - Reduce emission intensity of GDP by 45% by 2030 (from 2005 levels).
- Drivers: Falling solar tariffs, PM-KUSUM, PM Surya Ghar, Production Linked Incentive (PLI) for solar modules, wind-solar hybrid policy, and aggressive capacity addition (30–40 GW RE annually in recent years).

- Challenges: Intermittency → need for large-scale battery energy storage systems (BESS), pumped hydro, and grid modernisation; land acquisition; transmission bottlenecks; and round-the-clock (RTC)/Firm & Dispatchable Renewable Energy (FDRE) supply.
- Outcome: RE share in generation is rising steadily (solar + wind already contributing significantly during daytime).

Stage 4: Green Hydrogen – Long-term Decarbonisation Lever

- **Definition:** Hydrogen produced via electrolysis of water using renewable electricity (zero carbon footprint).
- **National Green Hydrogen Mission (NGHM)** – Launched January 2023:
 - Outlay: ₹19,744 crore (up to 2029–30).
 - Target: 5 Million Metric Tonnes (MMT) per annum green hydrogen production capacity by 2030.
 - Supporting RE capacity: ~125 GW.
 - Expected investment: > ₹8 lakh crore; 6 lakh jobs; ~50 MMT CO₂ emission reduction annually.
- **Key Components:**
 - Strategic Interventions for Green Hydrogen Transition (SIGHT) – incentives for electrolyser manufacturing and green hydrogen production.
 - Demand creation in hard-to-abate sectors (fertiliser/ammonia, refining, steel via Direct Reduced Iron, heavy transport, shipping).
 - Green Hydrogen Hubs, certification scheme, standards for green ammonia and green methanol.
- **Status (2026):** Early stage — ~8,000 tonnes per annum capacity commissioned; significant electrolyser manufacturing capacity awarded (~3,000 MW); green ammonia offtake agreements signed with fertiliser plants. Commercial scale still limited due to high costs.
- **Role:** Decarbonises sectors where direct electrification is difficult; positions India as a potential global exporter of green hydrogen and derivatives.

INDIA'S STRATEGIC PETROLEUM RESERVES AND ENERGY SECURITY

Current Status of Strategic Petroleum Reserves

India's Strategic Petroleum Reserves (SPRs) provide coverage of about **9.5 days of net oil imports**, while state-run oil companies hold crude oil and petroleum products storage equivalent to **64.5 days of net imports**. This brings the country's total national storage capacity for crude and petroleum products to roughly **74 days of net imports**.

SPR Phase I Capacity (5.33 million metric tons):

Location	Capacity
Visakhapatnam	1.33 million MT
Mangalore	1.5 million MT
Padur (Karnataka)	2.5 million MT

SPR Phase II Expansion (additional 6.5 million MT):

Location	Capacity
Chandikhol (Odisha)	4 million MT
Padur (Karnataka)	2.5 million MT

Additional SPR Facilities: ONGC is constructing a new 1.75 million MT SPR facility in Mangalore at a cost of \$1.6 billion.

International Comparison:

- Japan: ~200 days of coverage
- South Korea: ~207 days of coverage
- India: ~74 days total (well short of IEA's 90-day minimum)

Challenges in Managing Energy Resources

- **Energy Security:** Ensuring a steady and secure supply of energy resources is a challenge, especially with India's dependence on energy imports.
- **Environmental Impact:** The use of fossil fuels for energy generation leads to environmental degradation, including air pollution and climate change.
- **Energy Access:** Despite progress, a significant population in India still lacks access to clean and reliable energy sources, particularly in rural areas.
- **Energy Efficiency:** There is a need to enhance energy efficiency in industrial processes and transportation to reduce energy consumption.

Strategies for Sustainable Energy Resource Management

- **Diversification:** Diversifying the energy mix by investing in renewables like solar, wind, and hydropower can reduce dependence on fossil fuels.
- **Energy Efficiency:** Promoting energy-efficient technologies and practices can reduce energy consumption and costs.
- **Investment in Research and Development:** Investing in research and development can lead to technological advancements in energy production and storage.
- **Energy Access:** Expanding access to clean and reliable energy sources in rural and underserved areas is essential for inclusive development.

Government Initiatives

National Green Hydrogen Mission (NGHM)

- **Launched:** January 2023 (Cabinet approval).
- **Nodal Ministry:** Ministry of New and Renewable Energy (MNRE).
- **Outlay:** ₹19,744 crore (up to 2029-30).
- **Overarching Objective:** Make India a global hub for production, usage, and export of Green Hydrogen and its derivatives. Contribute to energy security, Atmanirbhar Bharat, and Net Zero 2070.
- **Key Targets by 2030:**
 - Green Hydrogen production capacity of at least **5 Million Metric Tonnes (MMT)** per annum.
 - Associated renewable energy capacity of **~125 GW**.
 - Attract investments of over **₹8 lakh crore**.

- Create ~6 lakh jobs.
- Reduce ~50 MMT of CO₂ emissions annually.
- **Major Components:**
 - **Strategic Interventions for Green Hydrogen Transition (SIGHT):** Incentives for electrolyser manufacturing and green hydrogen production (largest component of outlay).
 - Pilot projects in steel, mobility, shipping, and other hard-to-abate sectors.
 - Green Hydrogen Hubs and Infrastructure development.
 - R&D, standards, and certification framework (Green Hydrogen Certification Scheme operationalised; standards notified for Green Ammonia and Green Methanol).
- **Status (mid-2026):**
 - ~8,000 tonnes per annum capacity commissioned.
 - Incentives awarded for ~8.62 lakh MTPA green hydrogen production and 3,000 MW electrolyser manufacturing capacity.
 - Long-term green ammonia offtake agreements signed (covering ~7.24 lakh MTPA for fertiliser plants).
 - Early commercial projects by companies (e.g., JSW, Indian Oil); Hydrogen Valleys and port hubs under development.
- **Significance:** Critical for decarbonising hard-to-abate sectors (steel, fertiliser, refining, heavy transport) where direct electrification is difficult. Positions India in the global clean energy value chain.

National Solar Mission (Jawaharlal Nehru National Solar Mission - JNNSM)

- **Launched:** January 2010 under the National Action Plan on Climate Change (NAPCC).
- **Objective:** Establish India as a global leader in solar energy by creating policy conditions for rapid diffusion of solar technology across the country.
- **Original Target:** 20 GW grid-connected solar by 2022.
- **Revised Target (2015):** 100 GW by 2022 (60 GW utility-scale + 40 GW rooftop).
- **Current Context:** Solar is the largest contributor to India's renewable capacity and a cornerstone of the 500 GW non-fossil target by 2030.
- **Status (as of June 2026):**
 - Installed solar capacity: **162.15 GW** (57-fold increase from 2.82 GW in 2013-14).
 - Breakdown: Utility-scale ~125.60 GW, Rooftop ~30.11 GW, Off-grid ~6.43 GW.
 - India ranks **3rd globally** in installed solar capacity.
 - 56 Solar Parks sanctioned (total capacity ~39,461 MW); significant capacity already commissioned.
- **Supporting Schemes:**
 - Development of Solar Parks and Ultra Mega Solar Power Projects.
 - Production Linked Incentive (PLI) Scheme for High Efficiency Solar PV Modules (to boost domestic manufacturing).
 - PM Surya Ghar: Muft Bijli Yojana (rooftop solar for households).
- **Impact:** Dramatic cost reduction of solar power, large-scale capacity addition, improved energy access, and major contribution to non-fossil capacity share. Key driver of India's renewable energy success story.

National Policy on Biofuels & E20 Ethanol Blending

- **National Policy on Biofuels:** First notified in 2009; comprehensively revised in 2018 (amended 2022). Aims to promote biofuels for energy security, reduced oil imports, lower emissions, and farmer income support. Expanded feedstock base beyond sugarcane to include surplus food grains, maize, damaged grains, and agricultural residues (for 2G ethanol).

- **Ethanol Blended Petrol (EBP) Programme:**
 - Progressive blending of ethanol with petrol.
 - **Key Milestone:** Target of **20% ethanol blending (E20)** advanced from 2030 to Ethanol Supply Year (ESY) 2025-26.
 - **Achievement:** E20 target achieved in December 2025 / early 2026 — **five years ahead of the original schedule.** All public sector OMC retail outlets now dispense E20 petrol.
- **Progress of Blending:**
 - 2013-14: ~1.5%
 - Gradual rise → ~12% (2022-23) → ~14.6% (2023-24) → ~19.2% (2024-25) → **20%** (2025-26).
- **Benefits Realised (since 2014-15):**
 - Foreign exchange savings: > ₹1.9 lakh crore.
 - Crude oil substitution: ~310+ lakh metric tonnes.
 - CO₂ emission reduction: ~930+ lakh metric tonnes.
 - Additional farmer income: > ₹1.6 lakh crore.
- **Current Stance:** No decision yet to go beyond E20 (e.g., to E25/E30). Any higher blend will be based on scientific studies, vehicle compatibility, and stakeholder consultation. New vehicles are E20-compatible; studies indicate manageable impact on mileage (3-5% in some cases) with overall environmental and economic gains.
- **Supporting Measures:** Pradhan Mantri JI-VAN Yojana for advanced (2G/3G) biofuels; expansion of distillery capacity and feedstock diversification.

Integration with Energy Transition These three initiatives form critical pillars of India's clean energy strategy:

- **National Solar Mission** → Rapid scale-up of renewable electricity.
- **National Biofuel Policy / E20** → Decarbonisation of the transport sector and reduced oil import dependence.
- **National Green Hydrogen Mission** → Long-term solution for hard-to-abate industrial sectors.

Together, they operationalise the shift from fossil-fuel dominance toward a diversified, low-carbon energy system while supporting energy security, farmer incomes, and industrial competitiveness.

PM-KUSUM (Pradhan Mantri Kisan Urja Suraksha evam Utthaan Mahabhiyan)

- **Launched:** 2019 (Ministry of New and Renewable Energy).
- **Objective:** De-dieselise the farm sector, provide energy security to farmers, enhance farmer income, and add decentralised solar capacity.
- **Three Components:**

Component	Description	Target
A	Decentralised grid-connected solar/renewable power plants (0.5-2 MW) on barren/fallow/agricultural land; farmers sell power to DISCOM	10,000 MW
B	Installation of standalone solar agriculture pumps (off-grid)	~13-14 lakh pumps
C	Solarisation of existing grid-connected agriculture pumps (Individual Pump Solarisation - IPS + Feeder Level Solarisation - FLS)	~35 lakh pumps

- **Central Financial Assistance:** Up to 30–50% subsidy (varies by component and state contribution); farmers' contribution typically 10–20%.
- **Status (as of June 2026):**
 - Component B (standalone pumps): Strong performance (~11.5 lakh installed against ~13 lakh target; ~88% achievement).
 - Component C (especially Feeder Level Solarisation): Moderate progress.
 - Component A: Slower (~1,200–1,700 MW installed against 10,000 MW target).
- **Impact:** Reduces diesel dependence and electricity subsidy burden on states, provides daytime reliable power to farmers, creates additional income through power sale, and contributes to decentralised renewable capacity. Scheme timeline extended; next phase (PM-KUSUM 2.0) under discussion with greater focus on feeder-level solarisation.

PM Surya Ghar: Muft Bijli Yojana

- **Launched:** February 2024.
- **Outlay:** ₹75,021 crore.
- **Objective:** Install rooftop solar systems in **1 crore** residential households by FY 2026–27; provide free electricity (up to 300 units/month) to beneficiaries and promote household-level clean energy adoption.
- **Key Features:**
 - Central Financial Assistance (CFA) for rooftop solar installation.
 - National online portal for easy registration and application.
 - Focus on residential consumers; net metering benefits.
- **Status (mid/late 2026):**
 - Over 1 crore households registered.
 - ~33–39+ lakh rooftop systems installed (capacity addition ~12–14 GW).
 - Expected to cross 50 lakh beneficiary households soon.
 - Gujarat, Uttar Pradesh, Maharashtra, Rajasthan, and Kerala among leading states.
- **Impact:** World's largest domestic rooftop solar programme. Empowers households with energy self-sufficiency, reduces electricity bills, lowers daytime peak demand on the grid, creates local employment, and accelerates distributed renewable energy growth. Complements the National Solar Mission by focusing on the residential segment.

Offshore Wind Energy Programme

- **Background:** National Offshore Wind Energy Policy notified in 2015. India has significant technical potential (~70 GW estimated, mainly off Gujarat and Tamil Nadu coasts).
- **Targets:**
 - 30 GW offshore wind capacity by 2030 (with bidding trajectory of ~37 GW seabed lease areas by 2029–30).
- **Key Developments:**
 - Offshore Wind Energy Lease Rules notified (2023).
 - Strategy papers and zone identification completed (eight zones each in Gujarat and Tamil Nadu).
 - Viability Gap Funding (VGF) scheme approved (~₹7,453 crore) for initial 1 GW capacity (split between Gujarat and Tamil Nadu) plus port infrastructure support.
 - Bureau of Indian Standards has aligned design standards with international norms.
- **Status (2026):**
 - No commercial offshore wind project commissioned yet.

- Earlier tenders (totalling 4.5 GW) cancelled in 2025 due to lack of bids.
- Fresh tenders expected in the second half of 2026 (possibly 500 MW × 2 or 1 GW).
- International cooperation ongoing (e.g., India-UK Offshore Wind Task Force).
- **Challenges:** High capital costs, underdeveloped supply chain, specialised installation vessels, higher levelised cost of energy compared to onshore wind/solar, and need for port infrastructure.
- **Significance:** Complements onshore wind and solar by providing higher capacity utilisation factors and more consistent generation. Critical for achieving higher renewable penetration and the 500 GW non-fossil target while utilising India's long coastline.

Overall Linkage These three schemes strengthen the renewable energy pillar of India's energy transition:

- **PM-KUSUM** → Agricultural sector decarbonisation + farmer empowerment.
- **PM Surya Ghar** → Residential rooftop solar scale-up.
- **Offshore Wind** → Diversification beyond onshore resources for higher-quality renewable power.

Together with the National Solar Mission and Green Hydrogen Mission, they form a comprehensive strategy for distributed as well as utility-scale clean energy deployment.

Conclusion

Energy resources are the backbone of India's socio-economic development. While they fuel growth and improve the quality of life, they also pose environmental challenges. Sustainable management of energy resources, diversification of the energy mix, and investment in clean and efficient technologies are essential for India's energy security and environmental sustainability. By addressing these challenges and leveraging its vast renewable energy potential, India can pave the way for a sustainable and energy-secure future.